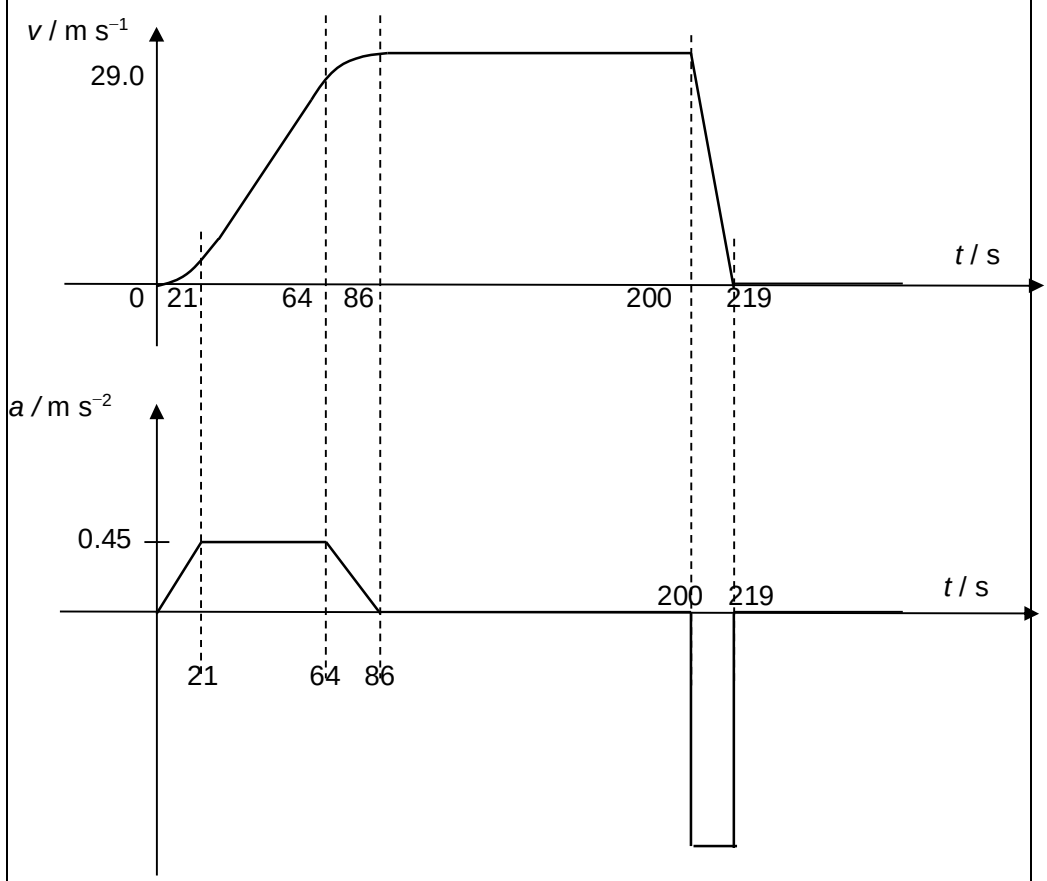


Section A

1	(a)	$\Delta v = \text{area under } a-t \text{ graph}$ $= \frac{1}{2} (21)(0.45) + (43)(0.45) + \frac{1}{2} (22)(0.45)$ $= 29$ $\langle a \rangle = \Delta v / \Delta t$ $= 29 / 200$ $= 0.15 \text{ (0.145) m s}^{-2}$	1 1
	(b) (i)	$v_{\text{max}} = \text{area under } a-t \text{ graph (from } t = 0 \text{ to } t = 86 \text{ s)}$ $= \frac{1}{2} (86 + 43) (0.45)$ $= 29.0 \text{ m s}^{-1}$	1 1
	(ii)	 The figure consists of two vertically stacked graphs sharing a common horizontal time axis labeled t / s. The top graph plots velocity $v / \text{m s}^{-1}$ against time. The curve starts at the origin (0,0), rises to a value of 29.0 at $t = 86$ s, remains constant at 29.0 until $t = 200$ s, and then decreases linearly to 0 at $t = 219$ s. Vertical dashed lines are drawn at $t = 21$, $t = 64$, $t = 86$, $t = 200$, and $t = 219$. The bottom graph plots acceleration $a / \text{m s}^{-2}$ against time. The acceleration is 0 until $t = 21$ s, then increases linearly to 0.45. It remains constant at 0.45 until $t = 64$ s, then decreases linearly to 0 at $t = 86$ s. From $t = 200$ s to $t = 219$ s, the acceleration is constant and negative, represented by a rectangular pulse below the time axis. Vertical dashed lines are drawn at $t = 21$, $t = 64$, $t = 86$, $t = 200$, and $t = 219$.	
2	(a)	[1] for the correct pair of forces	